INCH POUND
MIL-DTL-2953D
w/AMENDMENT 1
13 September 2018
SUPERSEDING
MIL-DTL-2953D
6 June 2018

# DETAIL SPECIFICATION

# STRAINERS, STEAM (SIZES 3 INCHES AND BELOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers basket-type steam strainers in sizes 3 inches and below.

1.2 Classification. Strainers covered by this specification are of the compositions, ratings, end styles, and sizes specified (see 1.2.1 through 1.2.4 and 6.2).

1.2.1 Composition code letter. The material of the strainer is identified as follows (see 3.1):

| Composition | Composition code letter |
|---|---|
| Composition A | A |
| Composition B | B |
| Composition D | D |

1.2.2 Rating code number. The pressure rating of the strainer is identified as follows:

| ASME B16.5 rating | Rating code number |
|---|---|
| Class 600 pound | 06 |
| Class 1500 pound | 15 |

1.2.3 End style code letter. The type of strainer end connections is identified as follows (see 3.2.5):

| End connection type | End style code letter |
|---|---|
| Flanged | F |
| Butt-Weld | B |
| Socket-Weld | S |

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

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1.2.4 Size code number. The strainer size is identified as follows (see 3.2.1):

| Nominal pipe size | Size code number |
|---|---|
| ½ | 08 |
| ¾ | 12 |
| 1 | 16 |
| 1¼ | 20 |
| 1½ | 24 |
| 2 | 32 |
| 2½ | 40 |
| 3 | 48 |

1.3 Part or identifying number (PIN). PINs to be used for strainers acquired to this specification are created as follows:

| **M** | **2953** | **-** | **X** | **XX** | **X** | **XX** |
|---|---|---|---|---|---|---|
| Prefix for Military Specification | Specification Number | - | Composition Code Letter (see 1.2.1) | Rating Code Number (see 1.2.2) | End Style Code Letter (see 1.2.3) | Size Code Number (see 1.2.4) |

Example: An example PIN created for a ½-inch, composition A, 600-pound rated, flanged strainer is M2953-A06F08.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-G-24716 - Gaskets, Metallic-Flexible Graphite, Spiral Wound

(Copies of these documents are available online at https://quicksearch.dla.mil.)



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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A193/A193M - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A194/A194M - Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High-Temperature Service, or Both

ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A696 - Standard Specification for Steel Bars, Carbon, Hot-Wrought or Cold-Finished, Special Quality, for Pressure Piping Components

(Copies of these documents are available online at www.astm.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.12 - Class 5 Interference – Fit Thread

ASME B16.5 - Steel Pipe Flanges and Flanged Fittings

ASME B16.11 - Forged Fittings, Socket-Welding and Threaded

ASME B16.25 - Buttwelding Ends

ASME Boiler and Pressure Vessel Code Section VIII

(Copies of these documents are available online at www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)



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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified in table I.

TABLE I. Materials.

| Name of part | Materials form | Composition A | Composition B | Composition D |
|---|---|---|---|---|
| Body | Casting | ASTM A217/A217M, grade WC9 | ASTM A217/A217M, grade WC6 | ASTM A216/A216M, grade WCB |
| Cover | Casting | ASTM A217/A217M, grade WC9 | ASTM A217/A217M, grade WC9 | ASTM A217/A217M, grade WC9 |
| | Bar stock | ------ | ------ | ASTM A696 |
| Basket screen | ------ | ASTM A240/A240M | ASTM A240/A240M | ASTM A240/A240M |
| Gaskets | Spiral wound | MIL-G-24716, type I | MIL-G-24716, type I | MIL-G-24716, type I |
| Studs for covers | Alloy steel for high temperature bolting | ASTM A193/A193M, grade B16 | ASTM A193/A193M, grade B16 | ASTM A193/A193M, grade B16 |
| Nuts for covers | Heavy semi-finished hexagon carbon and alloy steel for high temperature bolting | ASTM A194/A194M, grade 4 | ASTM A194/A194M, grade 4 | ASTM A194/A194M, grades 2, 2H, and 4 |

3.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Design and construction.

3.2.1 Sizes. Strainers shall be furnished in nominal pipe sizes of ½, ¾, 1, 1¼, 1½, 2, 2½, or 3, as specified (see 1.2 and 6.2).

3.2.2 End connections. Strainers shall have flanged ends in accordance with ASME B16.5 and table I, butt-weld ends in accordance with ASME B16.25, or socket weld ends in accordance with ASME B16.11, as specified (see 1.2 and 6.2). The design of the strainer nozzle cross-section, and the area between the end connection and the strainer body, shall be such that the connecting piping (considering maximum size and schedule to be used) will reach yield stress prior to the strainer nozzle cross-section. Face-to-face dimensions for flanges shall be in accordance with table II.

TABLE II. Flange face-to-face dimensions.

| ASME B16.5 rating | Strainer size (inches) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | ½ | ¾ | 1 | 1¼ | 1½ | 2 | 2½ | 3 |
| Class 600 | 5⅝ | 6⅜ | 7⅜ | 8⅜ | 9⅞ | 11⅞ | 13⅜ | 14⅜ |
| Class 1500 | 11 | 11¼ | 11½ | 13 | 14¼ | 17¼ | 20 | 23 |



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3.2.3 Covers. Covers shall be flanged and bolted. Covers shall not be fitted with any connection for a blow valve. Wye type strainer covers shall be provided with a boss to facilitate attachment of a blow down or drain connection. The following features shall be incorporated:

a. Cover alignment shall be obtained by body guiding, that is, a close tolerance fit between the cover and strainer body. The cover shall not be aligned by means of the cover fasteners.

b. Flange joint design shall assure proper compression of type I gaskets, in accordance with MIL-G-24716, when body and cover flanges are drawn up metal-to-metal.

c. Cover flanges shall be secured by any of the following:

(1) Through bolts threaded the entire length and fitted with a nut on each end. Threads on the bolts and nuts shall be class 2 fit.

(2) Tap end studs with interference class 5 fit at tap end and a class 2 fit at nut end. Interference shall be in accordance with ASME B1.12.

(3) Cap screws in accordance with B-7, class 2 fit of ASTM A193/A193M. Allen head type cap screws shall not be used.

d. Threaded fasteners, quantity and size, shall comply with ASME Boiler and Pressure Vessel Code, Section VIII, Rules for Bolted Flange Connections with Ring Type Gaskets.

3.2.4 Basket screen. The free area through the basket screen shall be at least 2 times the area of the inlet. Holes shall be either punched or drilled with all burrs removed. Holes shall be 0.020 inch in diameter and equally spaced over the basket. Spacing of holes on centers shall be approximately twice the diameter of the hole or greater. Metal thickness of basket shall be 0.015 inch minimum. Wire mesh shall not be used.

3.2.5 Strainer body. The design of all strainers shall provide that steam shall pass through the basket and shall be directed axially into the inside of the basket so that any foreign material intercepted by the strainer may be withdrawn by removal of the cover. The basket shall be installed in such a manner as to provide for easy removal. Each strainer shall have distinctly cast or stamped on one side of the body the size, the trademark of the manufacturer, and the pressure rating of the strainer. The walls of the bodies shall be curved surfaces; no flat surfaces shall be permitted. They shall be ribbed as necessary to provide strength. The low point of the strainer body shall be provided with a boss to facilitate installation of a drain connection by welding. In wye type strainers the boss on the strainer cover is considered to fulfill this requirement.

3.3 Pressure temperature. Pressure temperature rating shall be in accordance with ASME B16.5, class 600 and class 1500.

3.4 Shock. Strainers furnished in accordance with this specification shall meet the requirements of MIL-DTL-901 when tested in accordance with 4.3.2.

3.5 Nondestructive testing. Strainers shall be 100 percent radiographically inspected in accordance with S9074-AR-GIB-010/278.

4. VERIFICATION

4.1 Sampling for examination. A random sample of strainers shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.1.1 Visual examination. Sample strainers shall be visually examined in accordance with table III. Strainers not meeting the requirements of table III shall be rejected.



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TABLE III. Visual examination requirements.

| Examination | Test paragraph | Requirement paragraph |
|---|---|---|
| Dimensions | 4.1.2 | 3.2.1, 3.2.2 |
| Materials | 4.1.3 | 3.1 |
| Design and construction | 4.1.3 | 3.2 |
| Pressure and temperature | 4.1.3 | 3.3 |

4.1.2 Dimensional examination. Sample strainers shall be examined to determine conformance with the dimensional requirements specified herein (see 3.2.1 and 3.2.2).

4.1.3 Design. Sample strainers shall be examined to determine conformance with the material (see 3.1), design and construction (see 3.2), and pressure and temperature (see 3.3) requirements specified herein.

4.3 Tests.

4.3.1 Shell tests. Each of the strainers shall be subjected to the pressures specified in table IV for the duration of 1 minute. The water temperature shall not exceed 100 °F. Any weeping, leakage, or permanent deformation shall be cause for rejection.

TABLE IV. Test pressures for strength and porosity.

| ASME B16.5 rating | Pressure ($lb/in^2$) | |
|---|---|---|
| | Composition A and B | Composition D |
| Class 600 | 2,250 | 2,225 |
| Class 1500 | 5,625 | 5,575 |

4.3.2 Shock. Equipment and test classification shall be grade A, hull/bulkhead mounted, class III, type A. Method of mounting shall comply with figure 1.



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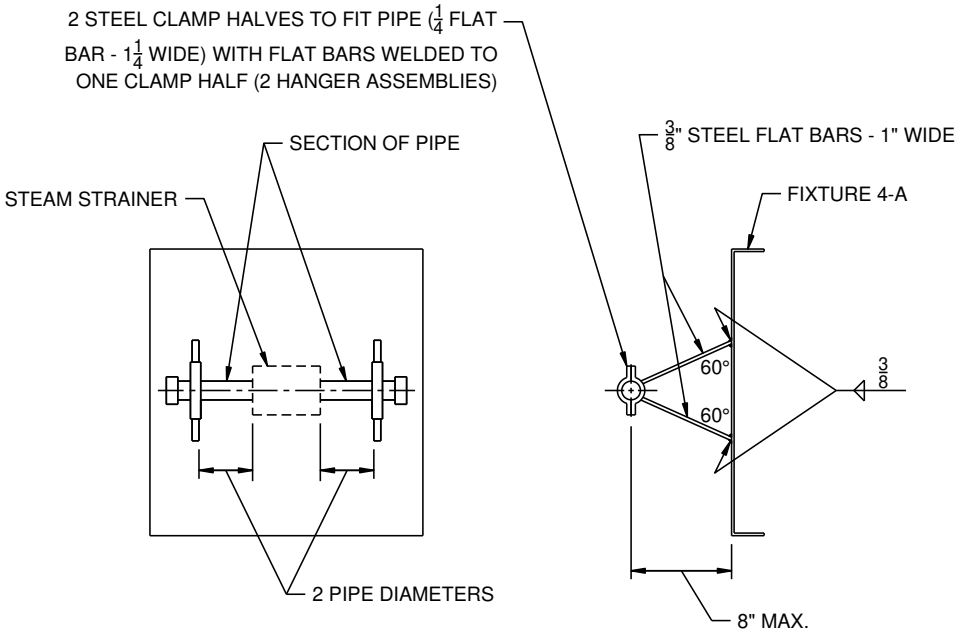



FIGURE 1. Straight tube, floating head, removable bundle, outside packed stuffing box.

4.3.2.1 Hydrostatic test. Strainer shall be hydrostatically pressurized with water to the cold rating specified in ASME B16.5. Any external leakage of the strainer or strainer/piping or visible damage to the strainer basket shall constitute failure.

4.3.3 Nondestructive tests. Strainers shall be 100 percent radiographically inspected in accordance with S9074-AR-GIB-010/278.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. Strainers covered by this specification are intended for steam and steam drain use only and have pressure-temperature ratings in accordance with ASME B16.5 except for maximum allowable temperatures. Strainers need to have service lives that are commensurate with the service life of a naval ship, and will be used in an environment that is subject to shock loadings associated with military service. The maximum temperature limitations are 1,050 °F for composition A strainers, 1,000 °F for composition B strainers, and 775 °F for composition D strainers. Current stock of strainers may be used until depleted; future requisitions for strainers should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. PIN (see 1.3).

c. Design and construction requirements (see 3.2.1 and 3.2.2).



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d. Quantity required (see 3.2.3).

e. AQL of 1.0 (see 4.1).

f. Packaging requirements (see 5.1).

6.3 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired for incorporation in the equipment and lose their separate identity when the equipment is shipped.

6.4 Subject term (key word) listing.

Basket

Body

Cover

Pressure

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH
DLA – CC

Preparing activity:
Navy – SH
(Project 4730-2018-087)

Review activities:
Army – AR
Navy – YD

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